Latest updates: https://dl.acm.org/doi/10.1145/3649902.3655648

PDF Download
3649902.3655648.pdf
28 February 2026
Total Citations: 1
Total Downloads: 294

Published: 04 June 2024

Citation in BibTeX format

ETRA '24: The 2024 Symposium on Eye Tracking Research and Applications
June 4 - 7, 2024
Glasgow, United Kingdom

Conference Sponsors:
SIGCHI
SIGGRAPH

ABSTRACT

Enhancing Cognitive Assessment: Integrating Hand and Eye Tracking in the Digital Trail-Making Test for Mild Cognitive Impairment

GUSTAVO JUANTORENA, University of Buenos Aires, Buenos Aires, Argentina

WALESKA BERRIOS, Italian Hospital of Buenos Aires, Buenos Aires, Argentina

MARÍA CECILIA FERNÁNDEZ, Italian Hospital of Buenos Aires, Buenos Aires, Argentina

AGUSTIN IBANEZ, Adolfo Ibáñez University, Santiago, RM, Chile

AGUSTÍN PETRONI, University of Gothenburg, Gothenburg, Vastra Gotaland, Sweden

JUAN E. KAMIENKOWSKI, University of Buenos Aires, Buenos Aires, Argentina

Open Access Support provided by:

Italian Hospital of Buenos Aires

Adolfo Ibáñez University

University of Buenos Aires

University of Gothenburg

Enhancing Cognitive Assessment: Integrating Hand and Eye Tracking in the Digital Trail-Making Test for Mild Cognitive Impairment

Gustavo E. Juantorena
Instituto de Ciencias de la Computación (FCEyN, UBA - CONICET)
Buenos Aires, Argentina
gjuantorena@gmail.com

Waleska Berríos
Neurology department, Hospital Italiano de Buenos Aires.
Buenos Aires, Argentina
waleska.berrios@hospitalitaliano.org.ar

María Cecilia Fernández
Neurology department, Hospital Italiano de Buenos Aires.
Buenos Aires, Argentina
mariacecilia.fernandez@hospitalitaliano.org.ar

Agustin Ibáñez
Trinity College Dublin (TCD)
Dublin, Ireland
Latin American Brain Health Institute (BrainLat)
Santiago, Chile
Global Brain Health Institute (GBHI), University of California San Francisco (UCSFA)
San Francisco, California, United States
agustin.ibanez@gbhi.org

Agustin Petroni
University of Gothenburg
Gothenburg, Sweden
agustinpetroni@gmail.com

Juan E. Kamienkowski
Instituto de Ciencias de la Computación (FCEyN, UBA - CONICET)
Buenos Aires, Argentina
juank@dc.uba.ar

ABSTRACT

We introduce a novel application of a computerized version of the Trail Making Test (c-TMT), designed to track hand and eye movements during multiple trials to tackle the main limitations of the classic pen-and-pencil version. Adults with mild cognitive impairment (MCI) and a matched control group were studied to obtain fine-grained metrics during the task. Statistically significant differences between groups were found in the scanpath length, while more traditional measures, such as time to complete trials, were not discriminative. We advocate for using new complementary measurements and digital tools in neuropsychology to improve the precision of neuropsychological tests.

CCS CONCEPTS

• **Applied computing** → **Health informatics**; • **Human-centered computing** → *Human computer interaction (HCI).*

KEYWORDS

Eye-tracking, Mouse tracking, Digital Neuropsychology, Mild Cognitive Impairment

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
ETRA '24, June 04–07, 2024, Glasgow, United Kingdom
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0607-3/24/06
https://doi.org/10.1145/3649902.3655648

ACM Reference Format:
Gustavo E. Juantorena, Waleska Berríos, María Cecilia Fernández, Agustin Ibáñez, Agustin Petroni, and Juan E. Kamienkowski. 2024. Enhancing Cognitive Assessment: Integrating Hand and Eye Tracking in the Digital Trail-Making Test for Mild Cognitive Impairment. In *2024 Symposium on Eye Tracking Research and Applications (ETRA '24), June 04–07, 2024, Glasgow, United Kingdom.* ACM, New York, NY, USA, 3 pages. https://doi.org/10.1145/3649902.3655648

1 INTRODUCTION

The Trail Making Test (TMT) is one of the most popular neuropsychological tests for executive functions (EFs) assessment [Bowie and Harvey 2006; Reitan 1958], used in both clinical evaluations and research studies. It consists of two sections, A and B. In Section A, participants are asked to connect a series of encircled items in numerical order using a pencil. In contrast, Section B requires individuals to strictly alternate between encircled numbers and letters in numerical and alphabetical order. The TMT has consistently demonstrated reliable results across diverse clinical populations. Several executive processes, including inhibitory control, working memory, and attention, are thought to influence TMT performance. In addition, it offers several strengths; it is easy for patients to understand, has a short administration time, and can be adapted for different cultural contexts, facilitating cross-cultural comparisons. However, it has important limitations. First, the underlying EFs articulated during the task are not well discriminated, which makes it a test with low specificity. Second, the pen-and-paper version presents one trial per condition, which introduces high variability. Third, only the total time is quantified, which does not allow for a detailed analysis. Fourth, it has a fixed spatial configuration per

condition. In a previous paper [Linari et al. 2022], we presented a computerized version of the TMT (c-TMT) that uses hand and eye tracking to get more precise data and new measures during the task. Here, we adapted this digital paradigm to test it in a population of patients with MCI and neurotypical controls. Our paradigm showed fine-grained measurements of eye and hand trajectories that were altered in patients. We show preliminary measures as candidates of digital EF markers in MCI patients.

2 METHODS

2.1 Participants

15 patients with mild cognitive impairment (age: 75.8 ± 7.0 years old) and 17 healthy controls matched for age (age: 72.5 ± 8.9 years old), were enrolled at the Hospital Italiano de Buenos Aires in Argentina. All participants had normal or corrected to normal vision and provided informed consent. All participants provided written informed consent. The protocol was approved by the IRB of Hospital Italiano de Buenos Aires (ID: 11543). The patients were diagnosed with MCI based on an interview, an informant report, a neuropsychological evaluation that showed a performance of -1.5 SD in at least two tests in one cognitive domain (attention, memory, language, executive functions, or visuospatial skills), and the ones having activities of daily living preserved [Jak et al. 2009]. A conservative criterion was used to determine the engagement of a cognitive domain, thus identifying the neuropsychological pattern as either uni- or multi-domain, and amnestic versus non-amnestic.

2.2 Stimuli

On each trial, 15 encircled items with numbers (TMT-A) or numbers and letters (TMT-B) were presented on the screen. The spatial distribution of the items was the same for all participants, but the order of the items was randomized across participants (see sample stimulus in Figure 1). For details, see [Linari et al. 2022].

2.3 Experimental Protocol

Stimuli were presented on a 24-inch screen (AOC G2460pf Freesync) at a viewing distance of 65 cm, subtending an angle of 29.3 degrees horizontally and 22.5 degrees vertically. Gaze locations were recorded from the right eye at a sampling rate of 500 Hz using an EyeLink 1000 plus (SR Research Ltd., Canada). A remote setup and a standard 9-point grid calibration were selected based on the characteristics of the population.

Participants had to click on a central cross to start each trial, after which targets appeared, and they had to connect them by passing them over. The mouse trajectory was drawn in real-time on the screen, and feedback was given when the elements were reached in order (the circles disappeared, but not the content with numbers or letters). Two practice and twenty experimental trials were presented, alternating TMT-A and TMT-B. Practice trials were discarded from the analysis. A trial ended when participants reached the last item or the time limit (25 seconds). The total time of the task did not exceed 12 minutes, including calibration.

All participants showed sufficient expertise using the mouse to avoid looking at it during the trial, which is a benefit over a tablet, albeit the tablet is more similar to the original paper and pencil.

Participants also completed other non-digital executive function tests, including MMSE, digit-symbol, paper-and-pencil TMT, and clock test, among others, as a comparison to validate our fine-grained digital metrics.

2.4 Eye-tracking Analysis

The eye-tracker software detected eye movements using the default cognitive parameters. Fixations that lasted less than 50 ms or exceeded 1000 ms were discarded. The fixation duration and the number of fixations per trial were then calculated separately by trial type (A or B) and group (MCI or control).

2.5 Hand-tracking Analysis

Based on the mouse trajectory (Figure 1), we calculated the proportion of trials where the mouse reached the last item following the correct sequence (Percentage of completion, PC) and the time spent (Response Time, RT).

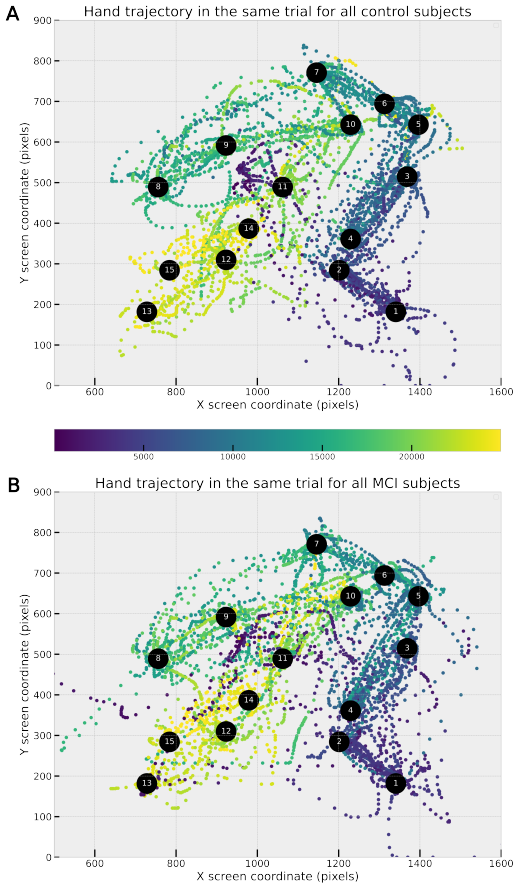

Figure 1: A,B. Examples of mouse trajectories for each subject and the same trial for the control (A) and MCI (B) groups. The color bar represents the temporal evolution during a trial.

2.6 Statistical Analyses

Two-way ANOVA was used to estimate the main effects of the subject group (MCI vs control), trial type (TMT-A vs TMT-B), and their interaction with the dependent variable. Analysis was performed using the Pingouin library [Vallat 2018] in Python.

3 RESULTS

Overall, the mouse trajectory and its variability looked similar between groups (Figure 1). This was confirmed by analyzing the RT (Figure 2.A) and PC (Figure 2.B), where differences were found for trial type (RT: F = 44.22, p = $9.9*10^{-9}$; PC: F = 32.70, p = $3.6*10^{-7}$), but not by group (RT: F = 0.26, p = 0.61; PC: F = 2.16, p = 0.15), nor a significant interaction (RT: F = 0.13, p = 0.72; PC: F = 1.00, p = 0.32). Exploring some broad variables from the eye movements, we found statistical differences for both trial type (F = 8.53, p = $4.91*10^{-3}$) and subject group (F = 5.16, p = 0.03) in the scanpath length (i.e. the number of fixations) (Figure 2 .D), but not for the interaction (F = $2.39*10^{-3}$, p = 0.96). Finally, there was no statistically difference in fixation duration for trial type (F = 1.70, p = 0.20), subject group (F = $9.1*10^{-4}$, p = 0.98), or interaction (F = 0.05, p = 0.82) (Figure 2.C).

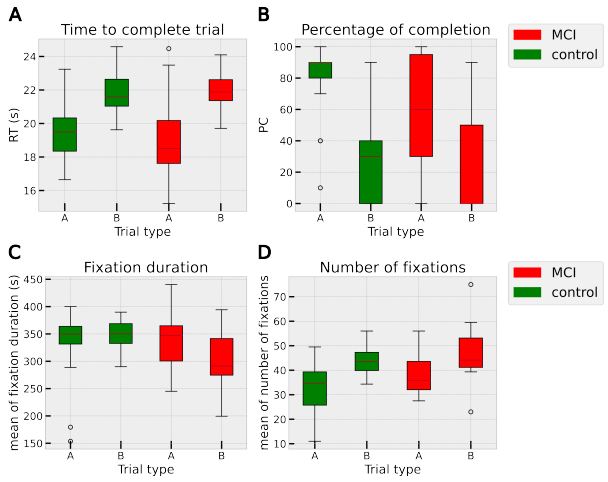

Figure 2: A. Time to connect the 15 items in order (RT) [mean ± std: TMT-A, MCI: 19.28±2.49; TMT-B, MCI: 21.94±1.38; TMT-A, control: 19.66±1.78; TMT-B, control: 21.95±1.53 s]. B. Percentage of completed trials (PC) [mean ± std: TMT-A, MCI: 60.00±36.06; TMT-B, MCI: 24.67±33.78; TMT-A, control: 77.53±24.05; TMT-B, control: 28.24±26.04] C. Fixation duration. [mean ± std: TMT-A, MCI: 335.04±55.23; TMT-B, MCI: 302.49±64.68; TMT-A, control: 329.40±66.46; TMT-B, control: 348.62±34.07 ms] D. Number of fixations per trial. [mean ± std: TMT-A, MCI: 38.50±8.76; TMT-B, MCI: 47.25±16.31; TMT-A, control: 32.90±10.18; TMT-B, control: 44.18±6.88].

4 CONCLUSIONS AND FUTURE WORK

By evaluating our c-TMT version in an MCI population, we observed significant differences with matched controls in an eye-tracking metric, the scanpath length (number of fixations) that went beyond classical measures failing to discriminate groups. Broader explorations of the combined hand and eye tracking measurements [Linari et al. 2022; Wölwer and Gaebel 2002] could provide a deeper understanding of EFs dysfunctions in the MCI population. Furthermore, metrics derived from this analysis can be correlated with standard (non-digital) indices from the same participants to validate our digital markers with these data. Our results suggest a promising path for applying this technology in digital neuropsychology [Armstrong et al. 2014; Germine et al. 2019; Montague et al. 2012; Schatz and Browndyke 2002; Smith et al. 2023].

ACKNOWLEDGMENTS

We thank Betsabe D. Leon Vallejos and M. Laura Chinen for their collaboration with the data acquisition. The research was supported by the Agencia Nacional de Promoción Científica y Tecnológica (PICT 2018-2699) and BrainLat (BL-SRGP2021-02).

REFERENCES

Kathleen A Armstrong, John L Semple, and Peter C Coyte. 2014. Replacing ambulatory surgical follow-up visits with mobile app home monitoring: modeling cost-effective scenarios. *Journal of medical Internet research* 16, 9 (2014), e3528.

Christopher R Bowie and Philip D Harvey. 2006. Administration and interpretation of the Trail Making Test. *Nature protocols* 1, 5 (2006), 2277–2281.

Laura Germine, Katharina Reinecke, and Naomi S Chaytor. 2019. Digital neuropsychology: Challenges and opportunities at the intersection of science and software. *The Clinical Neuropsychologist* 33, 2 (2019), 271–286.

Amy J Jak, Mark W Bondi, Lisa Delano-Wood, Christina Wierenga, Jody Corey-Bloom, David P Salmon, and Dean C Delis. 2009. Quantification of five neuropsychological approaches to defining mild cognitive impairment. *The American Journal of Geriatric Psychiatry* 17, 5 (2009), 368–375.

Ignacio Linari, Gustavo E Juantorena, Agustín Ibáñez, Agustín Petroni, and Juan E Kamienkowski. 2022. Unveiling Trail Making Test: visual and manual trajectories indexing multiple executive processes. *Scientific Reports* 12, 1 (2022), 14265.

P Read Montague, Raymond J Dolan, Karl J Friston, and Peter Dayan. 2012. Computational psychiatry. *Trends in cognitive sciences* 16, 1 (2012), 72–80.

Ralph M Reitan. 1958. Validity of the Trail Making Test as an indicator of organic brain damage. *Perceptual and motor skills* 8, 3 (1958), 271–276.

Philip Schatz and Jeffrey Browndyke. 2002. Applications of computer-based neuropsychological assessment. *The Journal of head trauma rehabilitation* 17, 5 (2002), 395–410.

Katharine A Smith, Charlotte Blease, Maria Faurholt-Jepsen, Joseph Firth, Tom Van Daele, Carmen Moreno, Per Carlbring, Ulrich W Ebner-Priemer, Nikolaos Koutsouleris, Heleen Riper, et al. 2023. Digital mental health: challenges and next steps. *BMJ Ment Health* 26, 1 (2023).

Raphael Vallat. 2018. Pingouin: statistics in Python. *Journal of Open Source Software* 3, 31 (2018), 1026. https://doi.org/10.21105/joss.01026

Wolfgang Wölwer and Wolfgang Gaebel. 2002. Impaired Trail-Making Test-B performance in patients with acute schizophrenia is related to inefficient sequencing of planning and acting. *Journal of psychiatric research* 36, 6 (2002), 407–416.